

Chapter 10

SUMMING UP

In Mathematics in 10 Lessons, Jerry P. King, Prometheus Books, 2009, p.369-380.

Long ago my professor told me what I was like. We talked in his campus office. He sat leaning back in his chair, feet up easy on his desk. I stood near the chalkboard, not quite at military attention. We had talked pure mathematics for an hour and then we discussed the progress, or lack of progress, of my doctoral dissertation. Now, as always, he was ending our discussion with a different kind of instruction.

Our weekly meetings consisted mostly of informal lectures delivered from me to him. I would write my new mathematics on the chalkboard. He would listen and criticize and offer suggestions. These sessions were intense and filled with apprehension and I measured the distance to my doctorate precisely by his reaction to the mathematics I put before him.

But, when my talk ended, he would invariably close the show with a lecture of his own, this one nontechnical and from him to me. The topics varied; you could not predict. My professor was a fine mathematician and from him I learned how to do mathematics. But, like many research mathematicians, his nontechnical interests were somewhat eccentric. Once he told me about Chinese agricultural methods for soybean growing. And once he described certain bizarre psychological characteristics he believed common to all academic administrators. Another time he explained in great detail his secret technique for swimming long distances underwater. But on this particular day he was being philosophical. This day, we had a rare dialogue.

“Mathematics graduate students,” he said, “are the brightest and the laziest students in any university.”

“Really?” I said, half pleased.

“Of course, there are exceptions to the ‘bright’ part,” he said looking me straight in the eye.

I was no longer even half pleased. “Why do you say mathematics graduate students are lazy?”

"Because they have discovered that to learn mathematics they need only learn a few fundamental things. The rest of mathematics comes from these few things."

"Why does that make them lazy?"

"Lazy by comparison," he said. "Look at other fields. A graduate student of literature cannot even begin work until he has read himself all the way around the walls of a library. And a chemist toils for months in a laboratory before his experiments even run properly, let alone produce anything new. Mathematics students do none of this."

"What must a mathematics student do?" I asked.

"Understand," my professor said.

"OK," I said, conceding. "Mathematics graduate students are lazy. But how do you know they are bright?"

"Because they've learned the secret."

"Secret?"

"Yes," he said. "They've learned that mathematics is pretty."

"How can you be sure they've learned it?"

"Because they are here," he said. "There is no other reason to study mathematics at this level."

"Why does that make them bright?"

"Because," he said, "they learned the secret by themselves. They took dozens of mathematics courses and they were told over and over that mathematics is useful. But no one told them it is beautiful. They learned that by themselves, one by one, all alone."

"But mathematics is useful."

"Certainly," he said. "Without mathematics there would be no technology, no modern world. But that's not why you do it. You do mathematics because it's pretty."

"And that's the secret they've learned?"

"Yes," he said. "They learned it alone and that makes them bright."

"How about me?" I asked. "I learned it too."

"Yes, but not by yourself."

"Who told me?"

"I did," he said, smiling. "Just now."

I did not then know what to make of the “bright but lazy” stuff my professor told me that day in his office. Nor do I now, all these years later. But I recognized even then that he was onto something no matter how casually he laid it out. And looking back I see that he came close to setting down for me the exact characteristics that separate mathematics from the other academic subjects and divide mathematicians from the rest of intellectual society.

But he did not set it down and years passed before I understood. I now think I do. I wish I could tell him I came to it myself the way his bright graduate students came alone to the secret of mathematics. But I’m not sure I did. Just as Moby Dick became part of each sea he swam, I am now touched by all I’ve done and by many of those I’ve met. My flanks are as scarred and barnacled as the white whale’s. I cannot always be certain of what is my own and what I’ve learned from others. But—from whatever source—I have over the passing years learned something about mathematics and about mathematicians.

My professor made three points in our long-ago dialogue:

- (1) *Beauty motivates mathematics*
- (2) *Mathematics flows from basic principles*
- (3) *Mathematics possesses utility.*

I made these three points at the beginning of our study. There, I asserted that points (1) and (2) characterize mathematicians in the sense that, among all the members of the intellectual world, only the mathematicians understand the true meaning of these two facts. Mathematicians also understand and appreciate point (3), as do many others. But it is not utility that motivates the creation of mathematics. My professor said it truly and well:

You do mathematics because it is pretty.

A phrase used by the contemporary mathematician Sherman Stein wonderfully condenses points (1), (2), and (3). Stein speaks of:

an appreciation of the beauty, extent, and vitality of mathematics.¹

Stein's *beauty* meshes exactly with point (1) above. I will interpret his *vitality* as "capacity to grow or develop." And I take *extent* to mean "the range over which something extends."

Thus the *vitality of mathematics* refers to the subject's ability to develop from a handful of embryonic principles into something as colossal and over-reaching as the great mathematical world itself. And the phrase the *extent of mathematics* suggests the enormity of this reach as it becomes the very language of Galileo's book of nature.

I wrote *The Art of Mathematics* to expose the concept of mathematical beauty. My specific purpose was to demonstrate the validity of the assertion: *beauty motivates mathematics*. I wanted to bring my professor's secret out into the sunlight. The present book considers mainly the *vitality* of mathematics. We began with only a few fundamental notions having to do with sets and counting. From these we moved, in an essentially self-contained manner, to some of the basic concepts of number theory, algebra, probability theory, and calculus. We saw mathematics grow from embryo to something reasonably adult.

But the concept of *extent* was always close at hand. We always kept before us the picture of the real world and the mathematical world side by side and the passing of mathematics from one to the other. Somehow the mathematical objects manage to ride an imaginary ether from their airy domain back into the real world where they predict and explain the behavior of concrete phenomena.

Perhaps this omnipresent picture yields an appropriate metaphor for the extent or the reach of mathematics. Give the word *reach* its old nautical meaning and then *mathematics becomes an abstract wind abeam with the world*.

THE AESTHETIC PRINCIPLE OF UTILITY

Physicist John Polkinghorne describes the essence of mathematics when he writes:

Mathematics is the abstract key that turns the lock of the physical universe.²

Here, in a single sentence, Polkinghorne sets down the quintessential nature of mathematics and of the universe, and he describes the relation between the two. Properly read, Polkinhorne's sentence says:

- (1) *Mathematics is abstract.*
- (2) *The universe is not abstract.*
- (3) *Mathematics explains and predicts physical phenomena.*

No one who has thought deeply about this doubts Polkinghorne's assertion. The long march of mathematicians and scientists from Archimedes' time until today has concurred. Science exists for the purpose of the formulation of theories that explain the past, help us understand the present, and in some small sense, predict the future. Scientists formulate their theories by a process that combines observation and extrapolation. They listen to the cosmos. And then they write their theories in mathematics because mathematics is the language that the cosmos speaks.

The process is always the same. A scientist paints, in the mathematical world, an abstract picture of real-world phenomena. This gives him a mathematical model of a portion of the real world. Next he does pure mathematics on the model. Then he extrapolates the new truths, yielded up by the mathematics, back to the real world. Polkinghorne says the process works. So do many others.

The physicist E. P. Wigner, you will remember, believes the effectiveness of mathematics is unreasonable. Martin Gardner, as we have seen, considers it only reasonable. Despite Gardner's brilliance and eminence, I concur with Wigner. Another who seems to concur is the physicist Freeman Dyson. In his book *Disturbing the Universe*, Dyson describes the pleasure he found in, for the first time, being allowed to do some serious laboratory experiments. Until this time, his work had been theoretical and mathematical. He watched an electron move on an oil drop, saw it move exactly as mathematics said it should move. He watched in wonder:

Why it is so, why the electron pays attention to our mathematics, is a mystery even Einstein could not fathom.

Never mind. Unreasonable or not, mathematics works.

Of course not every one thinks so. An old college pal had in his youth an interest in games of chance. He asked me one day about coin tosses.

“What is the probability a coin will land heads 100 times in a row?”

“If the coin is fair,” I said, “the probability is $\left(\frac{1}{2}\right)^{100}$.”

“That’s pretty small, isn’t it?”

“Tiny,” I said. “ $\left(\frac{1}{2}\right)^{100}$ is virtually zero. In decimal form there will be about 30 zeros following the decimal point.”

“Then it’s not likely to get 100 straight heads.”

“Not likely,” I said.

“OK,” he said. “But suppose this unlikely event has happened. Suppose the coin has landed heads 100 straight times. What’s the probability it comes up heads on the next toss?”

“One half,” I said.

“What?” he said.

“One half,” I repeated. “I’m assuming the coin tosses are independent of each other. If so, then the probability the coin lands heads on the 101st toss, given that it has landed heads on the first 100 tosses, is exactly $\frac{1}{2}$.”

“That’s crazy,” he said. “The coin surely remembers the many times it has landed heads. When the string gets long enough, it switches to tails.”

“How do you know when this will happen?” I asked.

“A real gambler knows.”

“The art of gambling, I suppose.”

“Exactly. But you should never gamble,” he told me. “You’ll never make money at it.”

And I never did. But, then, neither did he.

In those days my pal knew little mathematics and I was not surprised at his skepticism. Among the cognoscente, however, there is no skepticism. Mathematics turns the lock.

What do you do, however, if mathematics works too well? Suppose, for example, two scientists want to study a certain real-world phenomenon. After a certain amount of observation and reflection, each scientist follows the applied mathematics process described in figure 16 of chapter 1. Scientist A builds a mathematical model and, independently, so does scientist B.

Let's denote the phenomenon under consideration by the symbol P . Let M_1 and M_2 stand, respectively, for the mathematical models of scientist A and scientist B.

Each scientist works through the analysis stage of figure 16. Thus each scientist produces a collection of mathematical calculations that reveals truths about the respective models that were not known at the outset. When this is done, each scientist goes back to the real world and, through experimentation and observation, determines whether or not the new truths agree with real-world observation. That is, each scientist determines whether or not the theoretical model agrees with real-world observation.

We have here a potentially perplexing situation. It is possible that both models agree with observation and yet the models themselves differ from one another, and perhaps differ radically. Moreover, it may be that model M_1 and model M_2 offer vastly different predictions about future behavior, or about nonobservable behavior, of phenomena P . Which model then do we choose? Which is correct?

An important example of this situation occurred not all that long ago. In my graduate student days there existed two highly regarded and competing cosmological theories. The first of these—the big bang theory—remains before us. In this theory, as you know, the universe begins with a single explosion, a great fireball known as the big bang. In this theory the entire cosmos is born all at once, at a single instant. The second theory—called the steady-state theory—suggested a radically different scenario. In the steady-state theory, the creation of matter occurs continuously. Rather than the explosive, point-singularity birth suggested by the big bang theory, steady-state proposed a universe in which matter is being constantly created. Select a liter of universe out there somewhere in the void. Watch it carefully. Sooner or later a hydrogen atom will pop into existence.

Obviously, the two theories are inconsistent. They cannot both hold. Yet for a considerable period, scientists were unable to determine with finality which was correct. Both theories—in their observable aspects—agreed with what the scientists could see and measure. Thus for many years we had under consideration the exact situation described above: two vastly differing models M_1 and M_2 purporting to explain nothing less than the creation of universal matter. And we could not by observation choose between them. Only around 1965—when the discovery of background cosmic radiation provided a faint echo of the great blast—was the steady-state theory discredited.

So, in the case of steady-state versus big bang, observational evidence finally appeared that allowed scientists to choose one model over the other. But what if no such evidence had come forth? How do you decide between M_1 and M_2 in the absence of discriminating observational evidence? It's easy. You choose the more beautiful of the two.

The race may not always go to the swift and the battle to the strong, but that's how you bet. And the more beautiful of the two models may not be correct, but that's how you choose. Others will tell you the same thing. Listen to two famous physicists, Paul Dirac and Werner Heisenberg. Dirac said:

It is more important to have beauty in one's equations than to have them fit experiment . . . ³

And Heisenberg:

If nature leads us to mathematical forms of great beauty and simplicity . . . we cannot help thinking that they are true, that they reveal a genuine feature of nature.⁴

These two scientists tell you to choose the loveliest model whenever you can. Now, the models themselves are made of mathematics. And the mathematics itself, as we have seen, is motivated by aesthetics. The highest praise one mathematician can give to another is to pronounce the other's work *elegant*. G. H. Hardy put it this way:

Beauty is the first test: there is no permanent place in the world for ugly mathematics.⁵

So, we have a situation where mathematicians create mathematics for aesthetic reasons and then essentially allow the nonlovely mathematics to slide away. Then the scientists build their models of reality using the resulting mathematics. And, in addition, they choose between competing models—when they must—with aesthetic judgments. Thus the concept of beauty plays both a primary and a secondary role in the application of mathematics to real-world problems: first in the mathematics itself and then in the evaluation of competing models of reality. Put all this together and a startling principle falls out.

The aesthetic principle of utility: The most beautiful theories are the most useful.

As for steady-state versus big bang, the above principle gives the winner ahead of observation. It follows from the mathematics of the steady-state theory that, if matter is constantly created, then the rate of creation can be computed. This rate turns out to be a certain positive real number. Let's denote this number by r . But why is the rate this particular value? Why r ? Why should the universe settle on this particular continuous rate of creation of matter? Perhaps it did. But if I have a choice, I'd prefer matter to be created all at once. It seems to me much prettier. And the existence of the big bang theory gives me a choice. Thus big bang is preferable to steady-state for aesthetic reasons.

Of course, there is another possibility. Steady-state gives us rate of creation r . Big bang gives us matter all at once, an infinite rate of creation. How about $r = 0$? This seems to me even lovelier. I prefer the following cosmological principle.

The principle of the static universe: The universe is infinitely old. Matter was never created; it has always existed.

Any physicist will tell you immediately that the static universe is not a tenable possibility. It fails in many ways to agree with observation. No doubt the physicists are correct. Still there is about the static principle a certain minimal elegance.

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The extent of mathematics then—like the motivation for mathematics—is deeply intertwined with aesthetics. The aesthetic principle says “beauty implies utility.” Thus Polkinghorne's abstract key is shaped by the mathematician's concept of beauty. Poets similarly shape their poems. A great poet, John Keats, put it this way:

“Beauty is truth, truth beauty”—that is all
Ye know on earth, and all ye need to know.

Academic administrators, in particular, and much of humankind, in general, tend to think of mathematics as some kind of science. Mathematics departments are traditionally housed within universities as part of a college, or at least a division, of science. This division ordinarily contains the traditional departments of physics, chemistry, biology, and mathematics. And if you go outside the university and stop one hundred random citizens and ask: "Is mathematics an art or is it a science?" you are likely to receive one hundred votes for science. Yet it is almost obvious that mathematics cannot be science. There are several reasons.

(1) *Science is inductive; mathematics is deductive.*

(Scientists conclude, from the observational consistency of past events, the likelihood of future occurrences. Mathematicians deduce mathematical truth by logical argument from known mathematical facts.)

(2) *A theory is a scientific theory only if it is possible to show it to be false.*

(In late afternoon with my office door closed and the blinds drawn so that no one can see or hear, I can wiggle my finger and a copy of *The Complete Shakespeare* floats off my bookcase, and lands softly on my desk, opened to act 1, scene 1 of *Macbeth*. This is a nice theory of motion—a book moves when I point my finger—but it is not a scientific theory since you cannot show it is to be false. It works only when nobody is looking.)

(3) *A theory is a mathematical theory only if it has been proven true.*

(A proven mathematical theory has a name; it is called a theorem.)

Mathematics—we know—is necessary for science. But mathematics is not science. These two subjects are like the two men on the stairway in the famous M. C. Escher print. They walk in the same direction with their feet on the same step. But one goes up, the other down. "Contact between them is impossible," says Escher, "for they live in different worlds."

It is possible that a particular area of science may itself be an art just as mathematics is art. Mathematics, however, is not science.

The art nearest mathematics is the art of poetry. We know they both deal in beauty. But they have other common characteristics. For example:

(a) *Poetry is scary; mathematics is scary.*

(Many veteran actors avoid Shakespeare because speaking in verse frightens them. And everyone knows of the ubiquitous anxiety caused by fear of mathematics.)

(b) *Poetry is difficult; mathematics is difficult.*

(Examples abound. Here are two—one from each subject:

(i) Shakespeare writes:

Sith every action that hath gone before,
Whereof we have record, trial did draw
Bias and thwart, not answering the aim
And that unbodied figure of the thought
That gave't surmised shape.

As it stands this is difficult to understand. And, even if I tell you that the lines are from *Troilus and Cressida* and the speaker is Agamemnon, the difficulty remains. *Poetry is difficult.*

(ii) Look again at the epsilon-delta definition of the notion of limit given back in chapter 8. Write the definition carefully on a 3-inch-by-5-inch card. Carry the card next to your heart. Read it thoughtfully several times each day. When you truly understand it, try your hand at writing its negative. That is, write a precise statement of *what it means for the definition to be false*. Then see how long it takes you to understand the new statement. It took me about a year. *Mathematics is difficult.*)

(c) *Poetry is not natural; mathematics is not natural.*

(Both poetry and mathematics use language that is heightened, rich, and condensed. People these days do not naturally speak or write in verse. Nor do they naturally speak or write mathematics. Both poetry and mathematics must be *learned* and a taste for them must be *developed*. It is a fallacy to believe that either poetry or mathematics can be made naturalistic.)

A mathematician sees beauty in abstraction. But he also sees reality. A differential equation is a mathematical object and thus is completely abstract, like all such objects. And there are plenty of them, at least an uncountable infinity. But, by selecting a particular equation from this enormous set, a mathematician can build a model and thus describe the motion of a falling raindrop. In the abstract equation, the mathematician sees the reality of the raindrop's motion.

A poet works the other way around. When Robert Frost wanted to describe a spiritual experience, he wrote about stopping his horse on a snowy evening in a dark wood. Shakespeare placed infinite space inside a nutshell—or would have, had Hamlet not had bad dreams. Similarly Blake, the most antirational of poets, wrote:

To see a World in a Grain of Sand
And a Heaven in a Wild Flower,
Hold Infinity in the palm of your hand
And eternity in an hour.⁶

In the poet's creative hands the abstract notions of religion, infinite space, heaven, and eternity boil down to considerations of dark woods, nutshells, wild flowers, and an hour of time.

Both the mathematician and the poet trade in beauty and in abstraction. But they come at them in reverse order.

A mathematician sees all concrete images embodied in abstract ideas,

while

a poet sees all abstract ideas embodied in concrete images.

What then is the difference between the mathematician and the poet? Fundamentally, not much. They are face-to-face likenesses of one another, mirror images. Mathematicians know this instinctively. A mathematician looks in a mirror and a poet looks back. In a newer world it will also work the other way around.